Statistics
Parameter Estimation, Hypothesis Testing, Regression

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Lecture 5: Quick Run-through of Bayesian Estimation

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Bayesian Estimation

- Typically, we have a belief about the parameter, expressed as a CDF (or PDF) for the parameter, say $f(\theta)$.
- i.e., **prior distribution** on θ is assumed given.
- After measurement, you have additional information. You can now change your belief based on the measurement using Baye's theorem.
- i.e., obtain a **posterior distribution** on θ given the data.

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Bayesian Estimation

- We can thus write

$$f(\theta \mid x_1, x_2, \ldots, x_n) = \frac{f(x_1, x_2, \ldots, x_n \mid \theta)\, f(\theta)}{\int_\theta f(x_1, x_2, \ldots, x_n \mid \theta)\, f(\theta)\, d\theta}$$

- Using the posterior distribution, we could Maximum a Posteriori (MAP) estimate

$$\hat{\theta} = \arg\max_\theta f(\theta \mid x_1, x_2, \ldots, x_n)$$

- Or we could have the minimum mean square estimate

$$\hat{\theta} = \mathsf{E}(\theta \mid x_1, x_2, \ldots, x_n) = \int_\theta f(\theta \mid x_1, x_2, \ldots, x_n)\, d\theta.$$

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Bayesian Estimate: Example

- Bernoulli with prior probabilitiess $\mathsf{Prob}(p = 0.3) = 0.5$ $\mathsf{Prob}(p = 0.5) = 0.2$ $\mathsf{Prob}(p = 0.7) = 0.3$. Determine the for the posterior probabilities based on five samples.
- Bernouilli with uniform prior.
- Gaussian with variance σ and a Gaussian prior on μ.

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Lecture 6

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Hypothesis Testing

- Not interested in explicitly estimating the parameters but in verifying a statistical hypotheses about them.
  - value of single parameter: (1) $\mu = 1$, (2) $\mu > 1$
  - values of a set of parameters: (1) $\mu = 10$, $\sigma > 1$
  - distribution of the population: uniform, normal, exponential, etc.
- It is a hypothesis because it is not known if it is true.
- Test method is as follows: Design a suitable random experiment and accept the hypothesis if the data from the experiment is *consistent* with the hypothesis.
- The notion of *consistent* needs to be made more precise.
- Accepting the hypothesis does not mean "claiming that the hypothesis is true;
- Rather, we are saying *experimental data appears to be consistent with the hypothesis*. Once again notion of 'appears to be' will be made more precise.

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Hypothesis Testing

- **Null Hypothesis,** denoted by H_0 is to state that the initial claim or prior belief is true.
  - If there is the claim of a better drug, H_0 will be to state that this is not better than the current drug. (cures a higher fraction, or has lesser side effects)
  - If a new vendor to procure components is being tried, the new vendor is not any better than the current. (rejection ratio of the components is not lower).
  - If a new curriculum is being tested, then ....
- There is insufficient evidence in the data to change our beliefs or the 'status quo.'
- The word null means "of no value, effect or consequence," which suggests that H_0 should be identified with the hypothesis of no change, no effect or no improvement.
- H_0 is tested against an alternate hypothesis H_a or H_1.

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Test Procedures

- Conduct the random experiment and obtain the samples.
- Define the **Test Statistic,** a function of the sample data on which the decision to accept or reject H_0 is based.
- Define a **Rejection Region,** the set of all test statistic values for which H_0 will be rejected.
- Immediate question: How to obtain the rejection region? We need to see what are the possible errors from our experiment and conclusion.
- **Type I error:** Rejecting H_0 when H_0 is indeed true. The probability of Type I error is usually denoted by α.
- **Type II error:** Not rejecting H_0 when H_0 is false. The probability of Type II error is usually denoted by β.

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Example 1

- A new curriculum will make our students happier. Currently, 25% of our students are satisfied with the curriculum.
- You have resources for 20 samples. Let p be the actual fraction of students that are satisfied with the new curriculum.
- $H_0 : p = 0.25$
  $H_1 : p > 0.25$
- Let N be the number of students satisfied with the new curriculum.
- Consider the rejection region $N \geq 8$.
- If H_0 is true then N is a binomial random variable with parameters 20 and 0.25.
- Thus α is the probability of a binomial random variable with parameters $(20, 0.25)$ being greater than 8, i.e., $\alpha = 0.102$.
- The value of β depends on the true value of p.
- For true $p = 0.3$, β is the probability of $N \leq 7$ for a binomial random variable with parameters $(20, 0.3)$. $\beta = 0.772$.

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Example 2

- The monthly expenditure on food of a household income is Gaussian with mean 7,500 and standard deviation 900. With large retail stores (supermarkets in malls), the mean expenditure will reduce but the standard deviation will remain the same. We will check the expenditure of 25 households.
- Let X be the expenses after the supermarkets enter the area.
- H_0 : X is Gaussian with $\mu = 7500$ and $\sigma = 900$.
  H_1 : X is Gaussian with $\mu < 7500$ and $\sigma = 900$
- Test statistic is the sample mean, $\hat{\mu}$, of the expenses. This has $\sigma = 900/5 = 180$.
- Consider the rejection region $\hat{\mu} < 7080$.
- $\alpha =$ is the probability of a Gaussian random variable with parameters $(7500, 180)$ being lesser than 7080; this corresponds $\mathsf{Prob}(\mathsf{Z} < -7/3)$, i.e., $\alpha = 0.01$.
- The value of β depends on the true value of μ.
- For true $\mu = 7200$, β is the probability of $\hat{\mu} \geq 7080$ for a Gaussian variable with parameters $(7200, 180)$, i.e., $\beta = \mathsf{Prob}(\mathsf{X} > 2/3) = 0.7486$.
- For true $\mu = 7000$, $\beta = 0.174$.

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- In example 1, change rejection region to ≥ 9.
  $\alpha = 0.041$, $\beta(0.3) = 0.887$, and $\beta(0.5) = 0.252$.
- In example 2, change rejection region to ≤ 7200.
  $\alpha = 0.05$, $\beta(7200) = 0.5$, and $\beta(7000) = 0.1335$.
- Summary: Decreasing size of rejection region to obtain a smaller value of α results in a larger value of β for a particular set of the parameters.

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Some more terminology

- Simple hypothesis: If true, H_0 completely specifies the population. Both the preceding examples are simple hypotheses.
- Composite hypothesis: Not a simple a hypothesis.
- *Rejection Region:* The set of outcomes for which H_0 is rejected.
- Probability of a Type I error, α, is also called the significance of the test.
- Probability of a Type II error depends on true value of parameter, i.e., β is a function of p and μ in the preceding examples.
- **Important: Objective is not to explicitly determine if H_0 is true. Rather it is to determine if its validity is consistent with experimental data.** x
- H_0 is rejected if experimental data are very unlikely if H_0 is true.
- $1 - \beta(p)$ (or $1 - \beta(\mu)$) is called the *power function* of the test.

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Testing for Mean of a Normal Population with Known Variance

- $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$.
- Test obtains $X_1, \ldots, X_n$ and computes the sample $\hat{\mu}$.
- A reasonable rejection region: $\{X_1, \ldots, X_n : |\hat{\mu} - \mu_0| > c\}$
- $\hat{\mu}$ is Gaussian with unknown mean μ and known standard deviation σ.
- If α is specified, choose c such that $\mathsf{Prob}(|\hat{\mu} - \mu_0| > c \mid \mu = \mu_0) = \alpha$;
  Let $Z := \frac{\hat{\mu} - \mu_0}{(\sigma/\sqrt{n})}$

$$
\begin{aligned}
\mathsf{Prob}(|\hat{\mu} - \mu_0| > c \mid \mu = \mu_0) &= \alpha; \\
\mathsf{Prob}\left(|Z| > \frac{c\sqrt{n}}{\sigma}\right) &= \alpha \\
2\mathsf{Prob}\left(Z > \frac{c\sqrt{n}}{\sigma}\right) &= \alpha \\
\mathsf{Prob}\left(Z > z_{\alpha/2}\right) &= \alpha/2 \\
\frac{c\sqrt{n}}{\sigma} &= z_{\alpha/2} \\
c &= \frac{z_{\alpha/2}\sigma}{\sqrt{n}}
\end{aligned}
$$

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Testing for Mean of a Normal Population with Known Variance

- Reject H_0 if $\frac{\sqrt{n}}{\sigma}|\hat{\mu} - \mu_0| > z_{\alpha/2}$
  Accept H_0 if $\frac{\sqrt{n}}{\sigma}|\hat{\mu} - \mu_0| \leq z_{\alpha/2}$
- A reasonable choice of α depends on the specifics of the situation, essentially the 'cost' of a Type I error. It will also depend on 'the conviction' about H_0. Strong beliefs would require strong data to 'overthrow' H_0.
- An alternative view:
  - Consider $v = \frac{\hat{\mu} - \mu_0}{(\sigma/\sqrt{n})}$
  - Let p be the probability that the unit normal exceeds v.
  - Now observe that p is *critical significance level* of the test, i.e, if $\alpha < p$, H_0 is accepted.
  - p is called the p-value of the test.
  - p is the probability, caclulated assuming that H_0 is true, of obtaining the value of the test statistic (e.g., sample mean) at least as contradictory to H_0 as the value calculated from the available sample.

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Example

- H_0 : X is normal with mean $\mu = 8$ and $\sigma = 2$
  H_1 : X is normal with mean $\mu \neq 8$ and $\sigma = 2$
- From five samples, $\hat{\mu} = 9.5$.
- Test statistic:
$$\frac{\sqrt{n}}{\sigma}|\hat{\mu} - \mu_0| = \frac{\sqrt{5}}{2} \times 1.5$$
- For 5% level of significance, i.e., $\alpha = 0.05$, $z_{0.025} = 1.96$.
- Since test statistic is less than $z_{\alpha/2}$ accept H_0 at 5% significance.
- If we 'loosen' the criterion, i.e., allow a higher Type I error, choose $\alpha = 0.1$, then $z_{0.05} = 1.645$ and H_0 is rejected.
- If $\hat{\mu} = 8.5$, then $v = 0.559$.
  $\mathsf{Prob}(|Z| > 0.559) = 2\mathsf{Prob}(Z > 0.559) = 2 \times 0.288 = 0.576$.
- For $\hat{\mu} = 11.5$, $p = 0.00005$.
- You should verify these by using the Normal distribution table.

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Type II Error

- We now obtain $\beta(\mu)$.
- Define $Z = \frac{\hat{\mu}-\mu}{(\sigma/\sqrt{n})}$. Of course Z is the unit normal if the $\hat{\mu}$ were Gaussian with mean μ and variance σ^2/m.

$$\begin{aligned}
\beta(\mu) &= \text{Prob}\left(-z_{\alpha/2} \leq \frac{\hat{\mu}-\mu_0}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) \\
&= \text{Prob}\left(-z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \leq \frac{\hat{\mu}-\mu_0-\mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}}\right) \\
&= \text{Prob}\left(-z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \leq Z - \frac{\mu_0}{\sigma/\sqrt{n}} \leq z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}}\right) \\
&= \text{Prob}\left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}} - z_{\alpha/2} \leq Z \leq \frac{\mu_0-\mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) \\
&= \Phi\left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) - \Phi\left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right)
\end{aligned}$$

- Here $\Phi(x)$ is the CDF of the unit normal, i.e.,

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{y^2}{2}}\, dx$$

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Type II Error and Number of Samples

- Rejection region is determined by α.
- $\beta(\mu)$ can be used to determine the number of samples that may be required.
- Assume a requirement that $\beta(\mu_1) = \beta$, i.e.,

$$\Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right) = \beta.$$

- Let $\mu_1 > \mu_0$; then

$$\frac{\mu_0 - \mu_1}{(\sigma/\sqrt{n})} - z_{\alpha/2} \leq -z_{\alpha/2}$$

- Since Φ is an increasing function,

$$\Phi\left(\frac{\mu_0 - \mu_1}{(\sigma/\sqrt{n})} - z_{\alpha/2}\right) \leq \Phi(-z_{\alpha/2}) = \alpha/2 \approx 0$$

$$\beta \approx \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) = \Phi(z_{-\beta})$$

$$-z_\beta = \frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2}$$

- Solve for n.

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Remarks

- For tests involving mean of a normal population with known variance, we use the sample mean from n iid samples.
- The test statistic $Z = \frac{\hat{\mu}-\mu_0}{(\sigma/\sqrt{n})}$ measures the deviation of the sample mean from μ_0 in "units of the standard deviation."
- H_0 is rejected if Z is sufficiently large in the direction pointed to by H_1.
- Connections can be drawn with "Interval Estimate" of the mean of a normal population.

$$\text{Prob}\left(\mu \in \left(\hat{\mu} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\ ,\ \hat{\mu} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)\right) \quad = \quad 1-\alpha$$

- Hence if $\mu = \mu_0$, then with probability $(1-\alpha)$ it will be in the interval $\left(\hat{\mu} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\ ,\ \hat{\mu} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$.
- This means that H_0 is rejected with significance level α if

$$\mu_0 \quad \notin \quad \left(\hat{\mu} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\ ,\ \hat{\mu} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right).$$

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Bernoulli Populations

- H_0: $p \leq p_0$ versus H_1: $p > p_0$.
- Random binary valued samples X_i. Sample statistic $X = \sum_{i=1}^{n} X_i$.

$$\text{Prob}(X \geq k) \quad = \quad \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

- This is an increasing function in p. And if H_0 is true, we will analyse for $p = p_0$. Thus we have

$$\text{Prob}(X \geq k) \quad = \quad \sum_{i=k}^{n} \binom{n}{i} p_0^i (1-p_0)^{n-i}$$

- Reasonable rejection region would be $X > K^*$ where

$$K^* \quad = \quad \min\{k : \sum_{i=k}^{n} \binom{n}{i} p_0^i (1-p_0)^{n-i} < \alpha\}$$

- It is easier to determine the p-value first. Let $X = x$.

$$p - \text{value} \quad = \quad \sum_{i=x}^{n} \binom{n}{i} p_0^i (1-p_0)^{n-i}$$

Accept for $\alpha < p$-value.

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Example

- A manufacturer wants to give a six year warranty on a product. Expectation is that 90% will last longer than six years. "Accelerated life testing" is performed on 20 samples and results obtained. Let X be the number that survived.
- H_0: $p = 0.9$ versus H_1: $p_1 < 0.9$
- If the desired significance is $\alpha = 0.05$, then we want

$$K^* = \min\{k : \sum_{i=k}^{20} \binom{20}{i} 0.9^i (1-0.9)^{20-i} < \alpha\}$$

- Can be seen that $K^* = 15$ and $\alpha = 0.043$.
- HW: Evaluate β for $p = 0.8$.

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Bernoulli Populations: Large Samples

- X/n is an unbiased estimator for p.
- And X is approximately normal with mean np and variance $np(1-p)$.
- Thus $\frac{X-np}{\sqrt{np(1-p)}}$ is approximately unit normal.
- And all the theory from the previous "part" applies!
- For example, reject H_0 if

$$\frac{X - np_0}{\sqrt{np_0(1-p_0)}} \geq z_\alpha$$

- What do we do when H_0 is simple, i.e., H_0: $p = p_0$ versus H_1: $p \neq p_0$?
- Homework!

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Testing Two Populations

- Motivation:

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Testing Two Populations

- Motivation:
  - Two candidate treatments for a disease; interested in fraction cured.
  - Two processes to manufacture a product; interested in fraction of defects.
  - Or closer to you—two coaching classes; interested in fraction of successes.
- Probability of success is p_i for population i.
- With independent sampling all successes are independent with appropriate probability.
- Unequal sample sizes—n_i from population i; and X_i successes from n_i samples of population i.
- X_i is a Binomial random variable with parameters (n_i, p_i).
- H_0: $p_1 = p_2$ versus H_1: $p_1 \neq p_2$.

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Testing Two Populations

- Also under H_0 the samples form a population of size $(n_1 + n_2)$. And the total number of successes is Binomial with parameters $(n_1 + n_2, p_1)$.
- Let $X_1 + X_2 = k$.
- Note that we do not know p_1 or p_2; we are just checking to see if they are equal.
- Thus, under H_0, we expect X_1/n_1 to be close to X_2/n_2.
- Obtain the conditional probability of X_1, given k

$$\text{Prob}(X_1 = i | X_1 + X_k) = \frac{\binom{n_1}{i}\binom{n_2}{k-i}}{\binom{n_1+n_2}{k}}$$

- This is hypergeometric with parameters (n_1, n_2, k).
- Reject H_0 if X_1 is too small or too large.
- Assume the test resulted in $X_1 = x_1$.
- Reject H_0 if $\text{Prob}(X \leq x_1) \leq \alpha/2$ or $\text{Prob}(X \geq x_1) \leq \alpha/2$ where X is hypergeometric with parameters (n_1, n_2, k).
- Homework: What is the p-value of the test.
- This is called the Fisher-Irwin test.

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One Sided H_1.

- Recall the examples
  H_0: $\mu = \mu_0$ versus H_1: $\mu_0 > \mu_1$.
- Rejection region will be $\hat{\mu} > c$.
- For a given α, we obtain the following

$$\begin{aligned}
\text{Prob}(\hat{\mu} - \mu_0 > c) &= \alpha \\
\text{Prob}\left(\frac{\hat{\mu} - \mu_0}{(\sigma/\sqrt{n})} > \frac{c}{(\sigma/\sqrt{n})}\right) &= \alpha \\
\text{Prob}\left(Z > \frac{c\sqrt{n}}{\sigma}\right) &= \alpha \\
z_\alpha &= \frac{c\sqrt{n}}{\sigma} \\
c &= \frac{z_\alpha \sigma}{\sqrt{n}}
\end{aligned}$$

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One Sided H_1.

- $\beta(\mu)$ can be obtained as before

$$\begin{aligned}
\beta(\mu) &= \text{Prob}\left(\hat{\mu} \leq \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}}\right) \\
&= \text{Prob}\left(\frac{\hat{\mu} - \mu}{(\sigma/\sqrt{n})} \leq \frac{\mu_0 - \mu}{(\sigma/\sqrt{n})} + z_\alpha\right) \\
&= \text{Prob}\left(Z \leq \frac{\mu_0 - \mu}{(\sigma/\sqrt{n})} + z_\alpha\right) \\
&= \Phi\left(\frac{\mu_0 - \mu}{(\sigma/\sqrt{n})} + z_\alpha\right)
\end{aligned}$$

- Observe that $\beta(\mu)$ is decreasing in μ and $\beta(\mu_0) = 1 - \alpha$.

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